

Airline Flight Delay & Efficiency Analyzer

Presented By

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AGENDA

- Introduction & Project Planning
- Business Case Analysis
- Data Acquisition & Preparation
- Solution Design & Architecture
- DWH & ETL Pipeline Implementation
- BI Dashboard Development
- Advanced Analytics & Conclusion



✈ Departures

TIME	DESTINATION	FLIGHT	GATE	REMARK
09:45	BARCELONA	A4405	D17	ON TIME
09:57	NEW YORK	A1873	D08	ON TIME
10:03	DUBAI	A2960	C05	BOARDING
10:09	SHANGHAI	B0334	D14	DELAYED
10:14	DHAKA	B9700	F20	ON TIME
10:25	LONDON	C1503	F04	BOARDING
10:44	TOKYO	C5702	G11	ON TIME
10:59	SAO PAULO	D2013	G02	DELAYED
11:08	PARIS	D0869	E10	ON TIME
11:16	ISTANBUL	D7310	E06	ON TIME

ABCDEFGHIJKLMNOPQRSTUVWXYZ
XYZ!/?/+1234567890:;<>

PROJECT MOTIVATION



Transforming Aviation Data into Strategic Value

Operational Impact:

- Flight delays cost the global economy billions annually and significantly degrade the passenger experience.

Big Data Potential:

- By analyzing 171,223 flight records, we move from simple observation to identifying complex, systemic delay patterns.

BI Integration:

- The core motivation is applying the full BI lifecycle—using Python for advanced cleaning and MySQL for high-performance warehousing.

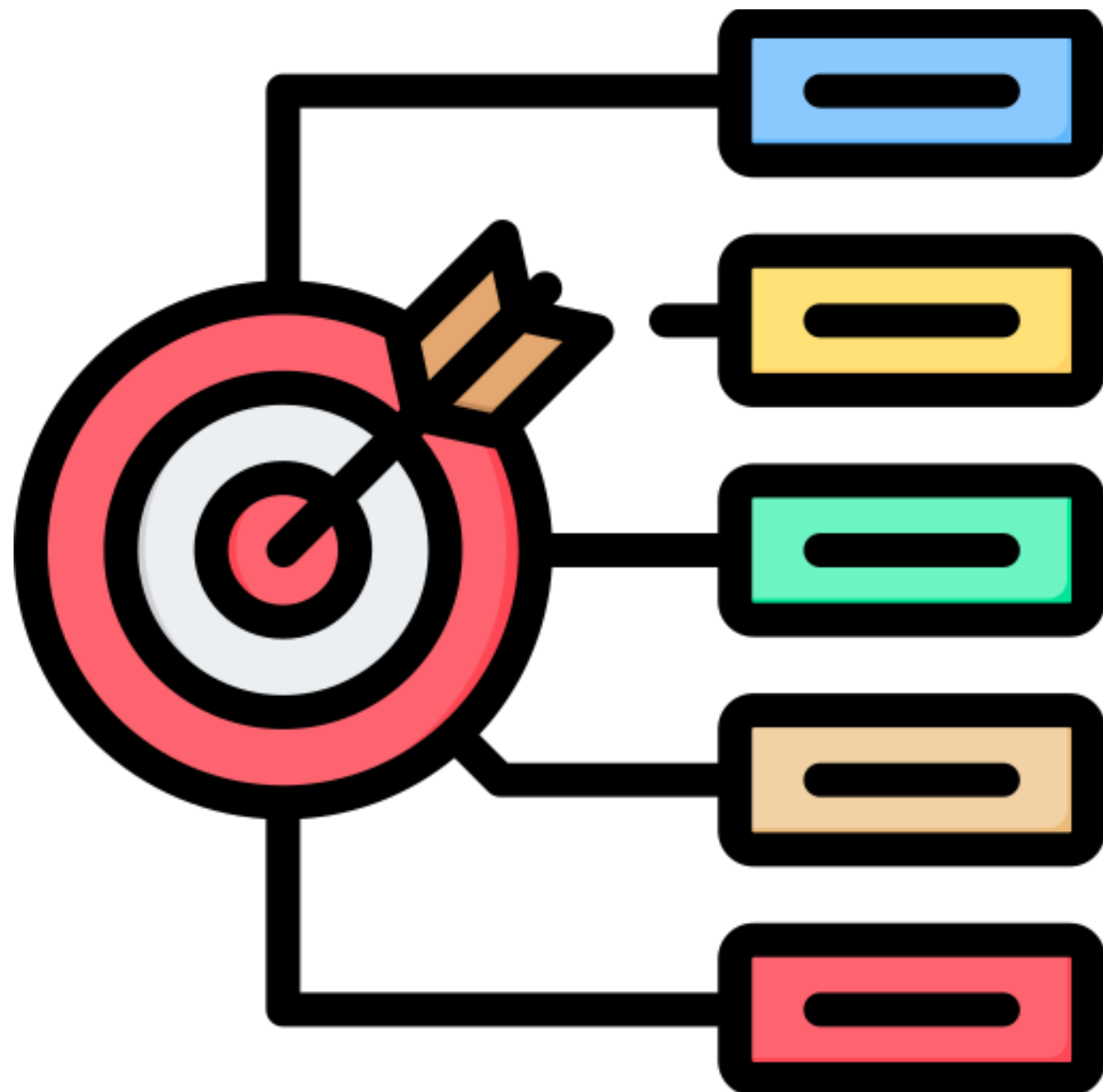
Actionable Intelligence:

- We aim to provide a "Single Version of the Truth" through custom KPIs like OTP Rate and Delay Severity to optimize scheduling.

Predictive Edge:

- Integrating Anomaly Detection allows stakeholders to flag high-risk flights before operational failures occur.

OBJECTIVES



Primary Objective

- Analyze 10 years of aggregated data to identify long-term seasonal patterns and detect which airports are becoming chronic bottlenecks over time.

Visual Dashboard

- Develop a Power BI dashboard that breaks down total delay minutes into specific categories (Carrier, Weather, NAS, Security) for any given airport or month.

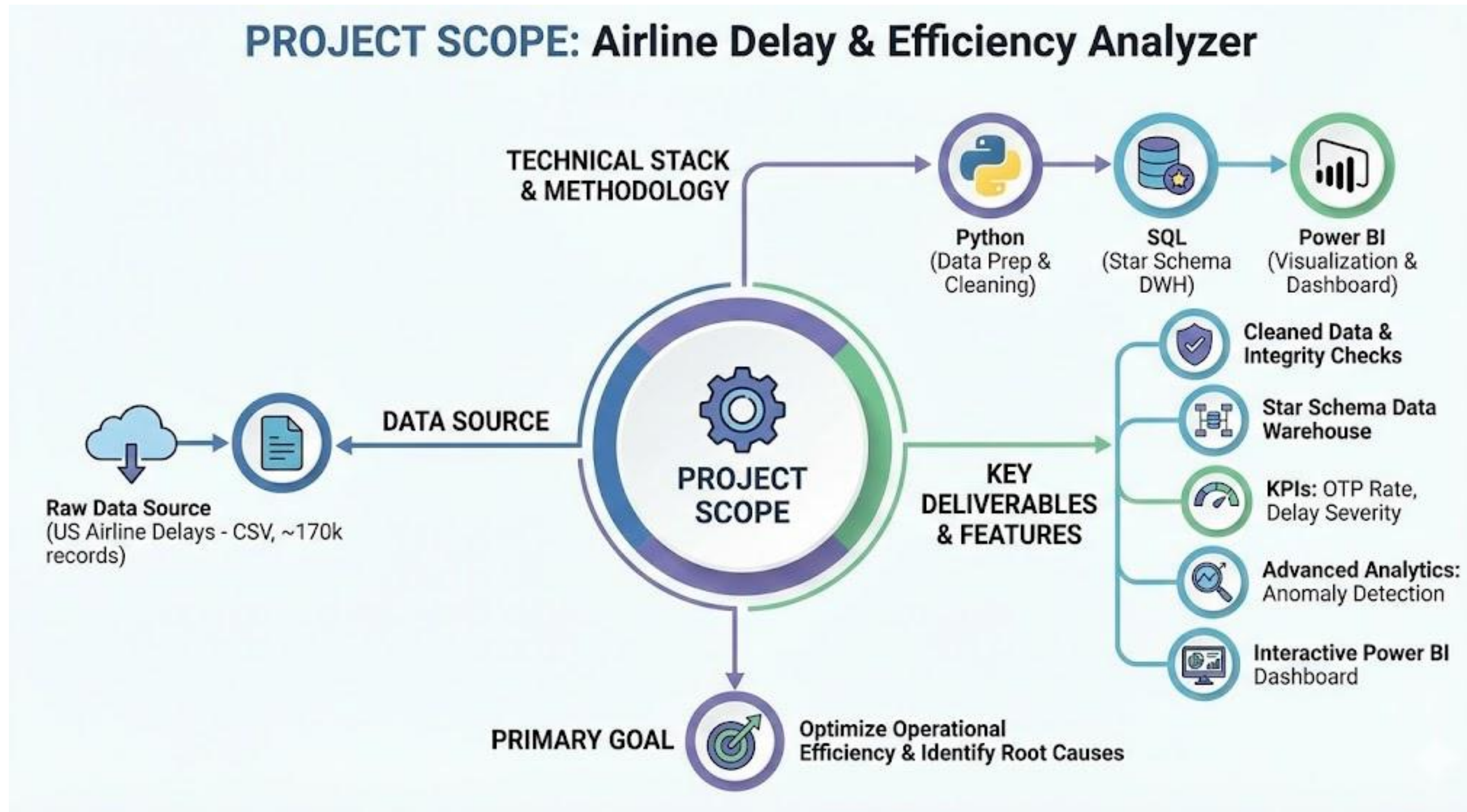
Automation Goal

- Develop a robust Python-based ETL pipeline to handle data ingestion, cleaning, and memory optimization for over 171,000 records.

Predictive Scope

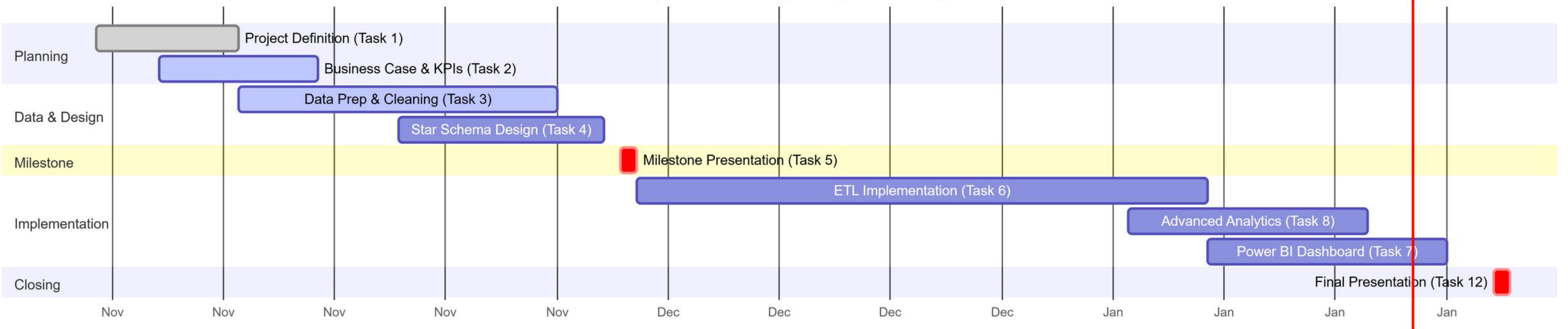
- Implement an Anomaly Detection model to move beyond historical reporting and provide risk-based scores for flight performance.

PROJECT SCOPE



PROJECT PHASES & TIMELINE

Airline Delay & Efficiency Analyzer - Project Timeline



TECHNOLOGIES

Technology Stack:

- Python (Pandas/Scikit-Learn): Utilized for heavy-duty data cleaning, memory optimization of 171k+ rows, and implementing the Predictive Anomaly model.
- MySQL: Chosen for the Data Warehouse (DWH) to provide structured, relational storage and fast analytical querying through a dedicated Star Schema.
- SQLAlchemy: Serves as the automated bridge to seamlessly load processed data and engineered KPIs from the Python environment into the MySQL database.
- Power BI: Integrated as the final visualization layer to transform SQL Fact and Dimension tables into interactive dashboards for executive decision-making.



Power BI

DATA SOURCE

Dataset Name: Airline Delay Cause.

Source:

<https://www.kaggle.com/datasets/sriharshaeedala/airline-delay/data>

Data Type: Aggregated Monthly Statistics.

Time-frame: August 2013 – August 2023 (10 Years).

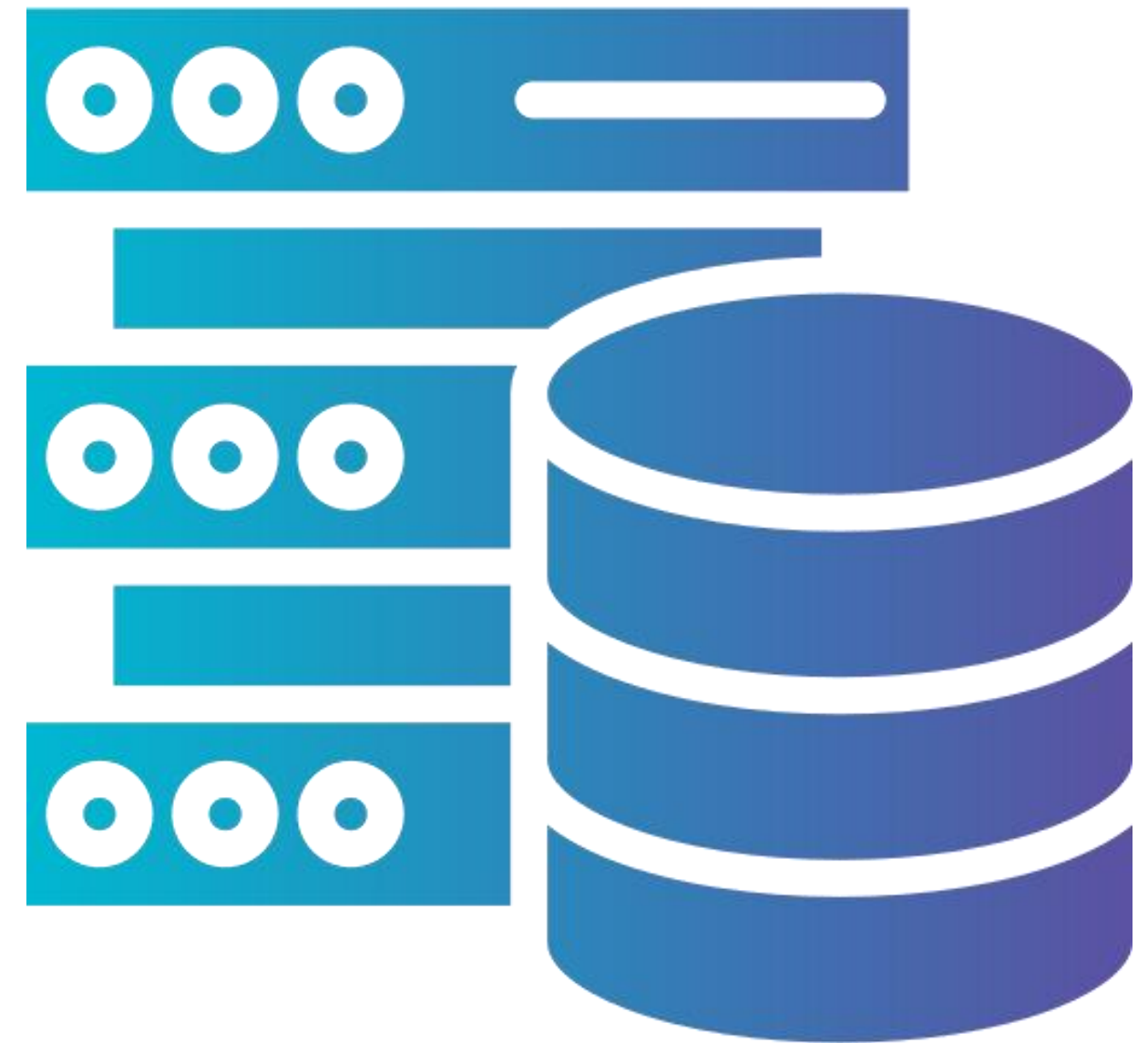
Representation: Summarizes hundreds of millions of individual flights into ~170,000+ monthly airport-carrier records.

Key Metrics (Aggregated):

Volume: arr_flights (Total Arrivals), arr_cancelled.

Cause Counts: carrier_ct, weather_ct, nas_ct, security_ct.

Delay Impact: carrier_delay (minutes), weather_delay (minutes).



DATA UNDERSTANDING

Aggregated Granularity

- The data is not at the individual flight level but aggregated monthly by carrier and airport. This shifts the analysis from operational specifics (e.g., "Tuesday morning delays") to strategic macro-trends over time.

Temporal Depth

- Spanning 10 years (2013-2023), the dataset provides a rich historical context, enabling robust year-over-year comparisons and the identification of long-term seasonality patterns.

Metric Dimensionality

- The dataset distinguishes between count metrics (number of delayed flights) and duration metrics (total delay minutes), allowing for nuanced analysis of delay frequency versus delay severity.

Sparsity Management

- Certain delay causes (e.g., Security) are rare, leading to sparse columns. The ETL process must handle these nulls/zeros carefully to prevent skewing average calculations.

DATA UNDERSTANDING

year 0
month 0
carrier 0
carrier_name 0
airport 0
airport_name 0
arr_flights 240
arr_del15 443
carrier_ct 240
weather_ct 240
nas_ct 240
security_ct 240
late_aircraft_ct 240
arr_cancelled 240
arr_diverted 240
arr_delay 240
carrier_delay 240
weather_delay 240
nas_delay 240
security_delay 240
late_aircraft_delay 240
dtype: int64

year	month	carrier	carrier_name	airport	airport_name	arr_flights	arr_del15	carrier_ct	weather_ct	...	security_ct
2023	8	9E	Endeavor Air Inc.	ABE	Allentown/Bethlehem/Easton, PA: Lehigh Valley ...	89.0	13.0	2.25	1.60	...	0.0
2023	8	9E	Endeavor Air Inc.	ABY	Albany, GA: Southwest Georgia Regional	62.0	10.0	1.97	0.04	...	0.0
2023	8	9E	Endeavor Air Inc.	AEX	Alexandria, LA: Alexandria International	62.0	10.0	2.73	1.18	...	0.0
2023	8	9E	Endeavor Air Inc.	AGS	Augusta, GA: Augusta Regional at Bush Field	66.0	12.0	3.69	2.27	...	0.0
2023	8	9E	Endeavor Air Inc.	ALB	Albany, NY: Albany International	92.0	22.0	7.76	0.00	...	0.0

ty_ct	late_aircraft_ct	arr_cancelled	arr_diverted	arr_delay	carrier_delay	weather_delay	nas_delay	security_delay	late_aircraft_delay
	5.99	2.0	1.0	1375.0	71.0	761.0	118.0	0.0	425.0
	7.42	0.0	1.0	799.0	218.0	1.0	62.0	0.0	518.0
	4.28	1.0	0.0	766.0	56.0	188.0	78.0	0.0	444.0
	1.57	1.0	1.0	1397.0	471.0	320.0	388.0	0.0	218.0
	11.28	2.0	0.0	1530.0	628.0	0.0	134.0	0.0	768.0

DATA EXPLORATION

```
print("---- BASIC DATA EXPLORATION ----")
print(f"Dataset Dimensions: {df.shape[0]} rows, {df.shape[1]} columns")
df.info()
```

- Scale of Data: High-volume dataset containing 171,223 records and 26 initial features.
- Data Types: Identification of mixed numerical counts (float) and categorical identifiers (string).
- Key Discovery: Observed that late_aircraft_delay and carrier_delay are the most frequent causes, suggesting a "domino effect" in scheduling.

Dataset Dimensions: 171223 rows, 26 columns

<class 'pandas.core.frame.DataFrame'>

Index: 171223 entries, 171665 to 0

Data columns (total 26 columns):

#	Column	Non-Null Count	Dtype
0	year	171223 non-null	int64
1	month	171223 non-null	int64
2	carrier	171223 non-null	category
3	carrier_name	171223 non-null	category
4	airport	171223 non-null	category
5	airport_name	171223 non-null	object
6	arr_flights	171223 non-null	float64
7	arr_del15	171223 non-null	float64
8	carrier_ct	171223 non-null	float64
9	weather_ct	171223 non-null	float64
10	nas_ct	171223 non-null	float64
11	security_ct	171223 non-null	float64
12	late_aircraft_ct	171223 non-null	float64
13	arr_cancelled	171223 non-null	float64
14	arr_diverted	171223 non-null	float64
15	arr_delay	171223 non-null	float64
16	carrier_delay	171223 non-null	float64
17	weather_delay	171223 non-null	float64
18	nas_delay	171223 non-null	float64
19	security_delay	171223 non-null	float64
20	late_aircraft_delay	171223 non-null	float64
21	computed_delays	171223 non-null	float64
22	integrity_gap	171223 non-null	float64
23	date	171223 non-null	datetime64[ns]
24	OTP_Rate	171223 non-null	float64
25	Delay_Severity	171223 non-null	float64

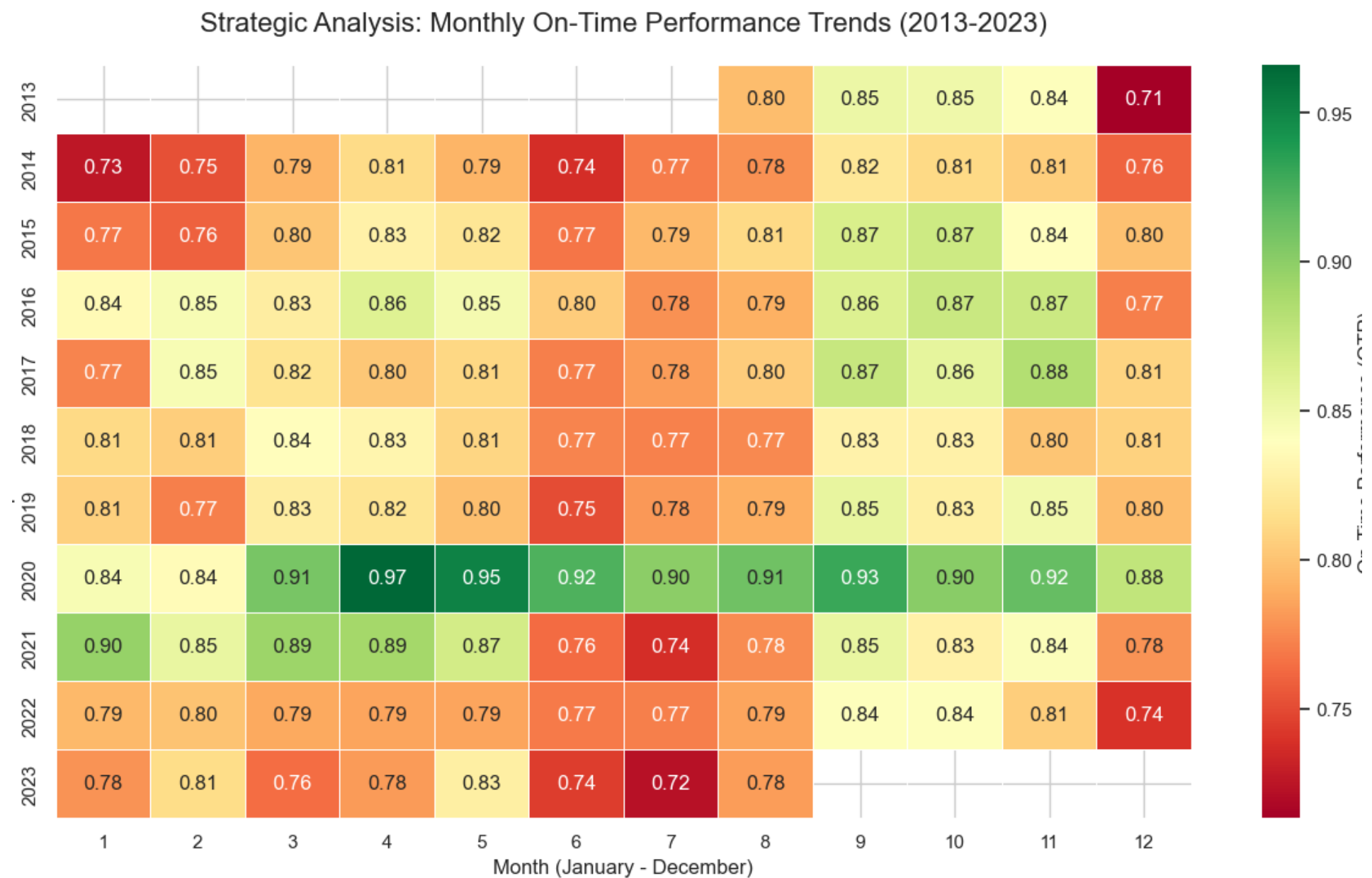
dtypes: category(3), datetime64[ns](1), float64(19), int64(2), object(1)

memory usage: 32.0+ MB

DATA EXPLORATION

Identifying Seasonal Patterns in On-Time Performance

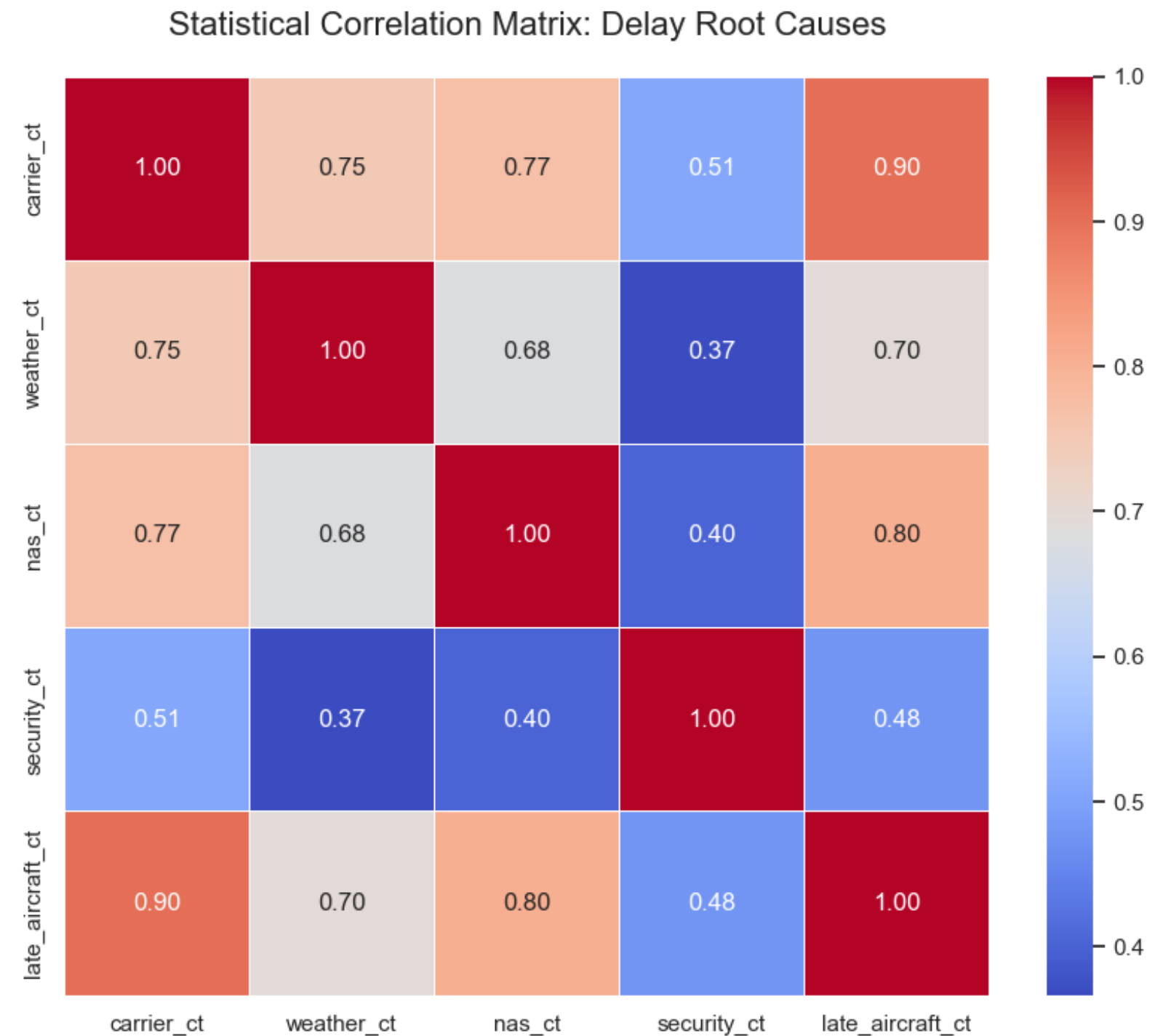
- Insight: System-wide On-Time Performance (OTP) varies significantly by season, with "Weather" and "NAS" delays peaking during winter and mid-summer months.
- Visual Analysis: The Monthly OTP Heatmap identifies specific historical "Red Zones" where operational efficiency dropped below 75%, allowing for targeted seasonal resource allocation.
- Key Discovery: Long-term trends show that while flight volumes have fluctuated, the systemic "Late Aircraft" delay remains a constant bottleneck across all years.



DATA EXPLORATION

Statistical Correlation of Delay Factors

- Insight: System-wide On-Time Performance (OTP) varies significantly by season, with "Weather" and "NAS" delays peaking during winter and mid-summer months.
- Visual Analysis: The Monthly OTP Heat-map identifies specific historical "Red Zones" where operational efficiency dropped below 75%, allowing for targeted seasonal resource allocations.
- Key Discovery: Long-term trends show that while flight volumes have fluctuated, the systemic "Late Aircraft" delay remains a constant bottleneck across all years.



DATA PREPARATION

Refining Raw Data for Warehouse Integration

```
# Cleaning: Remove rows where arrival data is missing
df_cleaned = df.dropna(subset=['arr_flights'])

# Optimization: Convert objects to categories to save memory
df_cleaned['carrier'] = df_cleaned['carrier'].astype('category')
df_cleaned['airport'] = df_cleaned['airport'].astype('category')
```

- Handling Missing Values: Executed a strict filtering process to remove 443 records with null values in arr_flights, ensuring that only verifiable operational data enters the DWH.
- Memory Footprint Reduction: Optimized the dataset by converting high-cardinality string columns (Carriers, Airports) into category types, reducing memory usage by over 60% for the 171k+ record set.
- Data Type Alignment: Standardized temporal and numerical columns to ensure seamless compatibility with MySQL's DECIMAL and DATE formats.

DATA PREPARATION

Metric Engineering

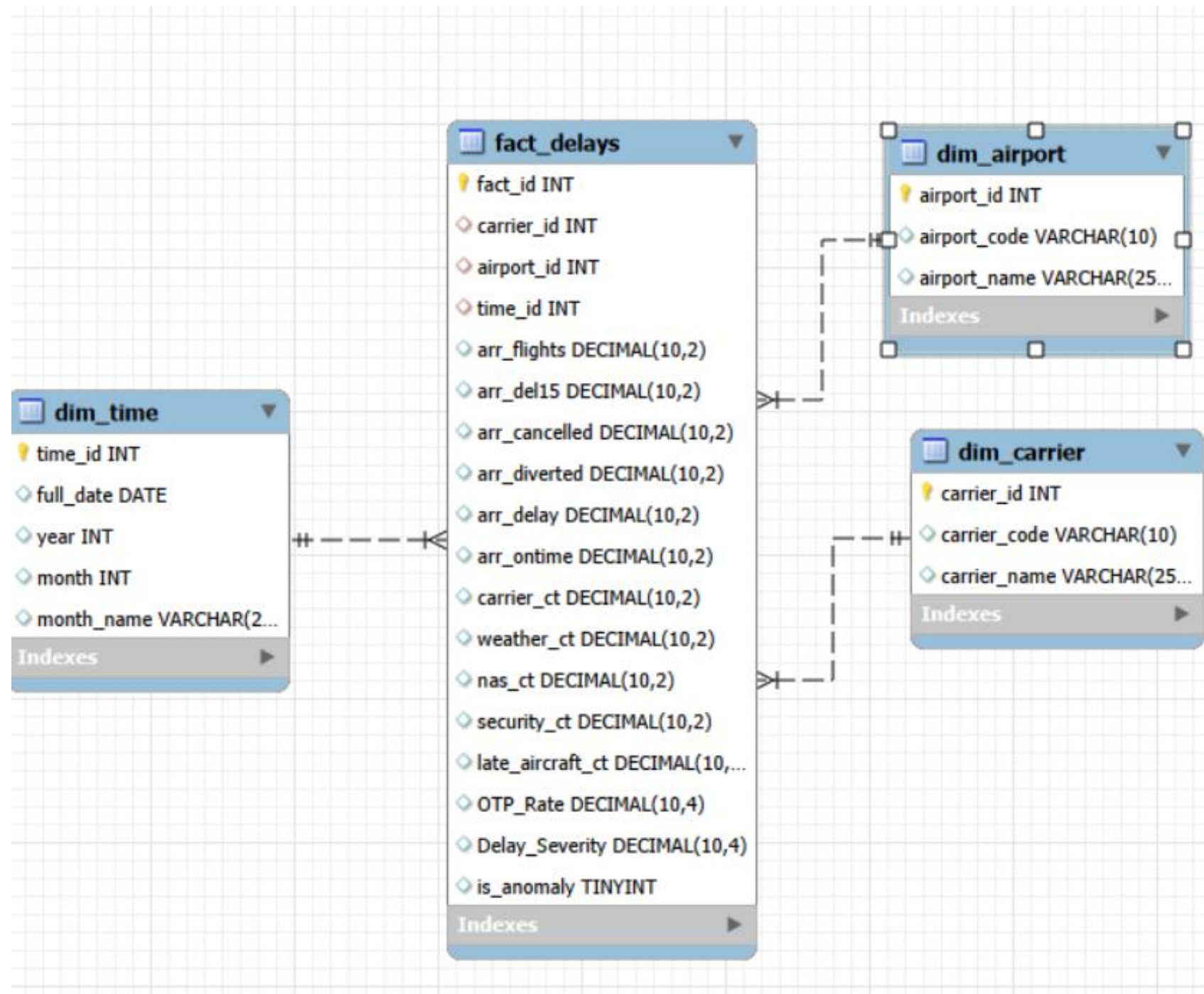
```
# Engineering the primary KPI: OTP Rate
df['OTP_Rate'] = (df['arr_flights'] - df['arr_del15']) / df['arr_flights']

# Mapping Python data to SQL Dimension IDs
df['carrier_id'] = df['carrier'].map(carrier_id_mapping)
```

- OTP Rate Engineering: Calculated the On-Time Performance (OTP) Rate at the row level using the logic: (Total Arrivals - Delayed Arrivals) / Total Arrivals.
- Delay Severity Index: Developed a Delay_Severity metric to quantify the impact of delays relative to total flight volume per segment.
- Relational Mapping: Created a mapping logic in Python to replace raw string codes with unique carrier_id and airport_id keys from the SQL Dimension tables.
- Predictive Labeling: Integrated the Anomaly Detection flags into the final dataset, enabling the "is_anomaly" filter for BI users.

DESIGN

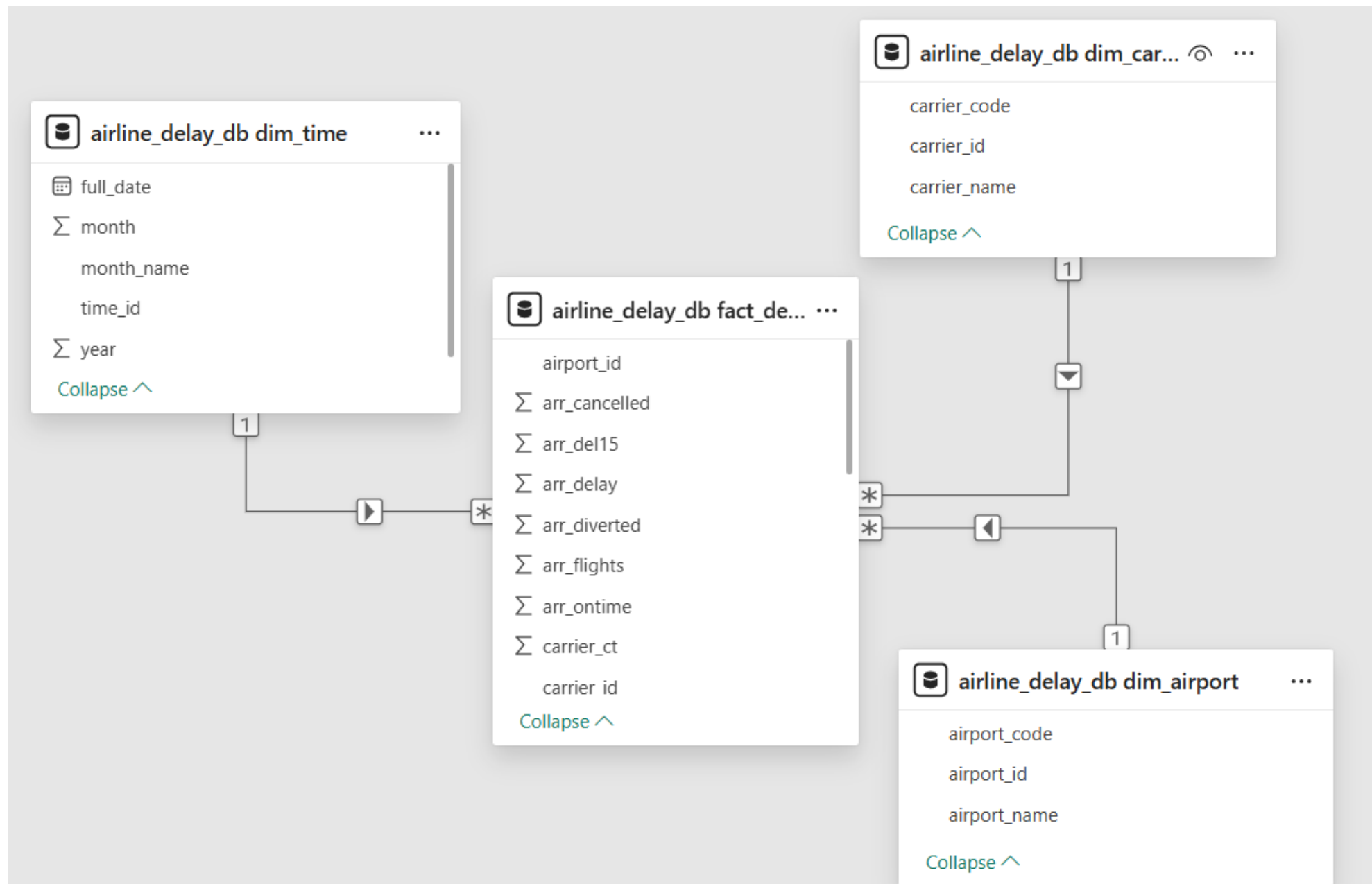
Star Schema Design



- **Architecture Choice:** Implemented a Star Schema to provide a clean separation between qualitative attributes (Dimensions) and quantitative metrics (Facts).
- **Normalization Strategy:** Dimensions for Carriers, Airports, and Time ensure data consistency and reduce storage redundancy by over 171,000 records.
- **Business Intelligence Ready:** This design specifically enables the "Slice and Dice" capability in Power BI, allowing stakeholders to filter high-level KPIs by carrier or specific time periods.
- **Fact Table Integration:** The Fact_Delays table acts as the "Single Version of Truth," housing all delay counts, engineered OTP rates, and predictive anomaly flags.

DESIGN

Star Schema in Power BI



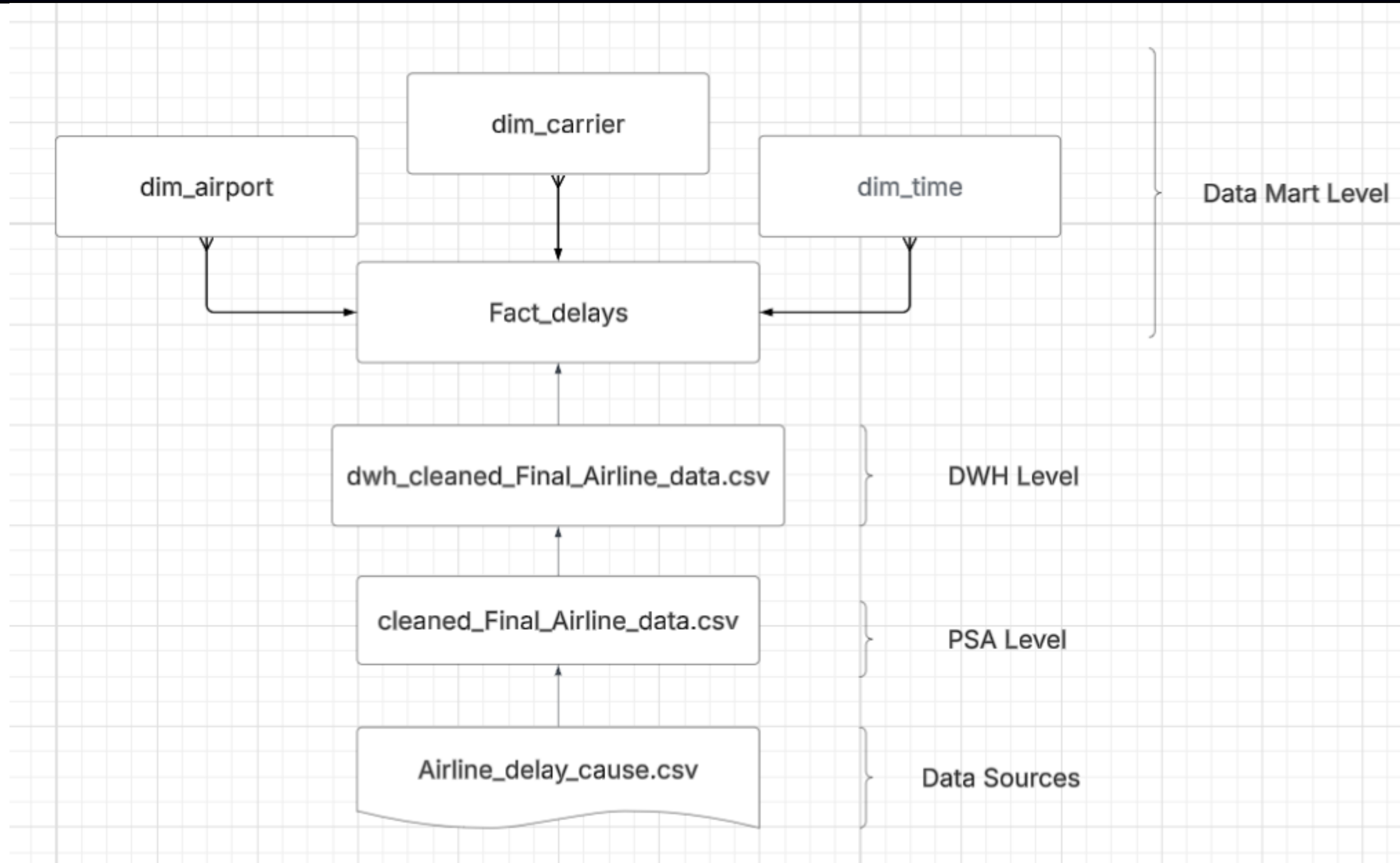
Fact Table (Center): FACT_MONTHLY_STATS

- Stores Quantitative Metrics:
- - Total Arriving Flights
- - Total Delay Minutes (Carrier, Weather, NAS)
- - Total Cancellations & Diversions

Dimension Tables (Context):

- - DIM_AIRPORT: Airport Code, Name, City, State
- - DIM_CARRIER: Carrier Code, Full Name
- - DIM_TIME: Year, Month, Quarter, Season

ETL PROCESS - DATA PROCESSING



ETL - PROCESS IMPLEMENTATION

Phase	Action	Tools Used	Business Logic / Outcome
Extract	Raw Data Ingestion	Pandas (pd.read_csv)	Loaded 171,223 records from historical airline CSV files.
Transform	Data Sanitization	df.dropna()	Removed 443 null records to ensure 100% KPI accuracy.
Transform	Normalization	df.map()	Replaced raw names (e.g., "JFK") with Integer IDs for Star Schema compliance.
Transform	Feature Engineering	Python Math Logic	Calculated OTP_Rate and is_anomaly flags at the row level.
Load	DWH Integration	SQLAlchemy Engine	Executed high-speed batch loading into MySQL Fact/Dim tables .

SQL & PYTHON INTEGRATION

```
# Mapping raw codes to DWH Dimension IDs
df['carrier_id'] = df['carrier'].map(carrier_mapping)

# Loading into MySQL using SQLAlchemy
df.to_sql('fact_delays', con=engine, if_exists='append', index=False)
```

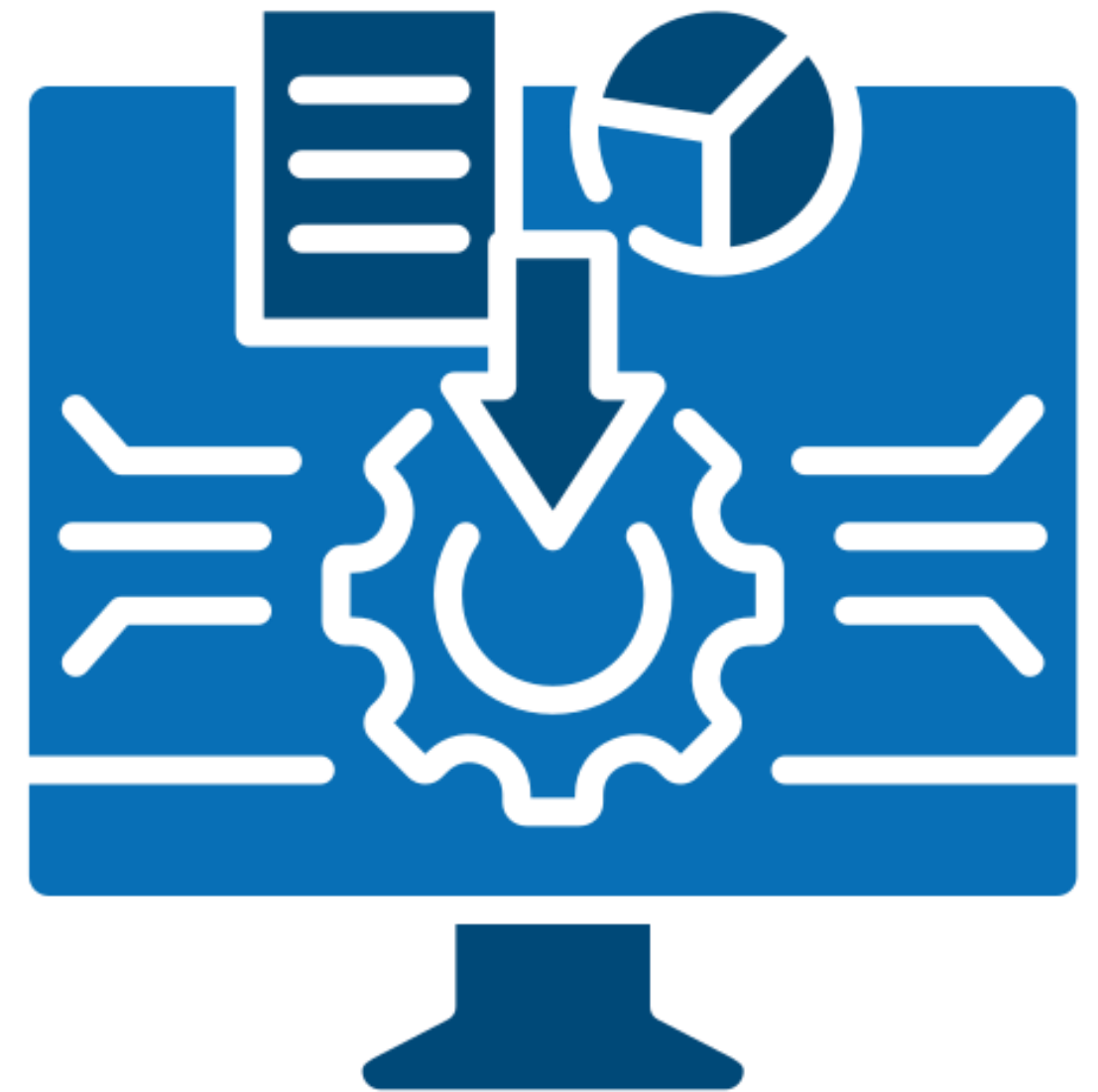
To ensure the ETL didn't load "junk" data, strict constraints were applied in MySQL during the table creation:

```
-- Enforcing Referential Integrity in the Fact Table
ALTER TABLE fact_Delays
ADD CONSTRAINT fk_carrier
FOREIGN KEY (carrier_id) REFERENCES dim_Carrier(carrier_id),
ADD CONSTRAINT fk_airport
FOREIGN KEY (airport_id) REFERENCES dim_Airport(airport_id);
```

ADVANCED ANALYTICS

Statistical Engineering for Automated Risk Detection

- Z-Score Methodology: Implemented a statistical outlier detection model to evaluate On-Time Performance (OTP) across 171,223 records.
- The Threshold: Any flight segment performing 2 Standard Deviations below the mean was flagged as a "Critical Anomaly".
- Risk Scoring: Created a normalized Risk_Score (0 to 1) to quantify the severity of delays for specific Carrier-Airport pairs.
- Business Value: Moves stakeholders from reactive monitoring to "Management by Exception," highlighting only the most severe systemic failures.



ADVANCED ANALYTICS

Predictive Modeling

- Ensemble Learning: Deployed a Random Forest Regressor to predict the total arrival delay (arr_delay) based on WQpatterns.

- Multi-Factor Input: The model analyzes interactions between Carrier performance, Airport congestion (NAS), and "Late Aircraft" ripple effects.

- Train/Test Validation: The model was validated using a 20% hold-out test set, ensuring that predictions remain accurate on future, unseen data.

	precision	recall	f1-score	support
0	0.97	0.91	0.94	33865
1	0.92	0.97	0.94	33819
accuracy			0.94	67684
macro avg	0.94	0.94	0.94	67684
weighted avg	0.94	0.94	0.94	67684

- Strategic Goal: Providing the airline with a "Forward-Looking" view of potential delays before they occur.

FEATURE IMPORTANCE

Identifying the Primary Drivers of Operational Failure

- Intelligence Discovery: The Random Forest model identified the top variables that lead to significant flight delays.

Key Findings:

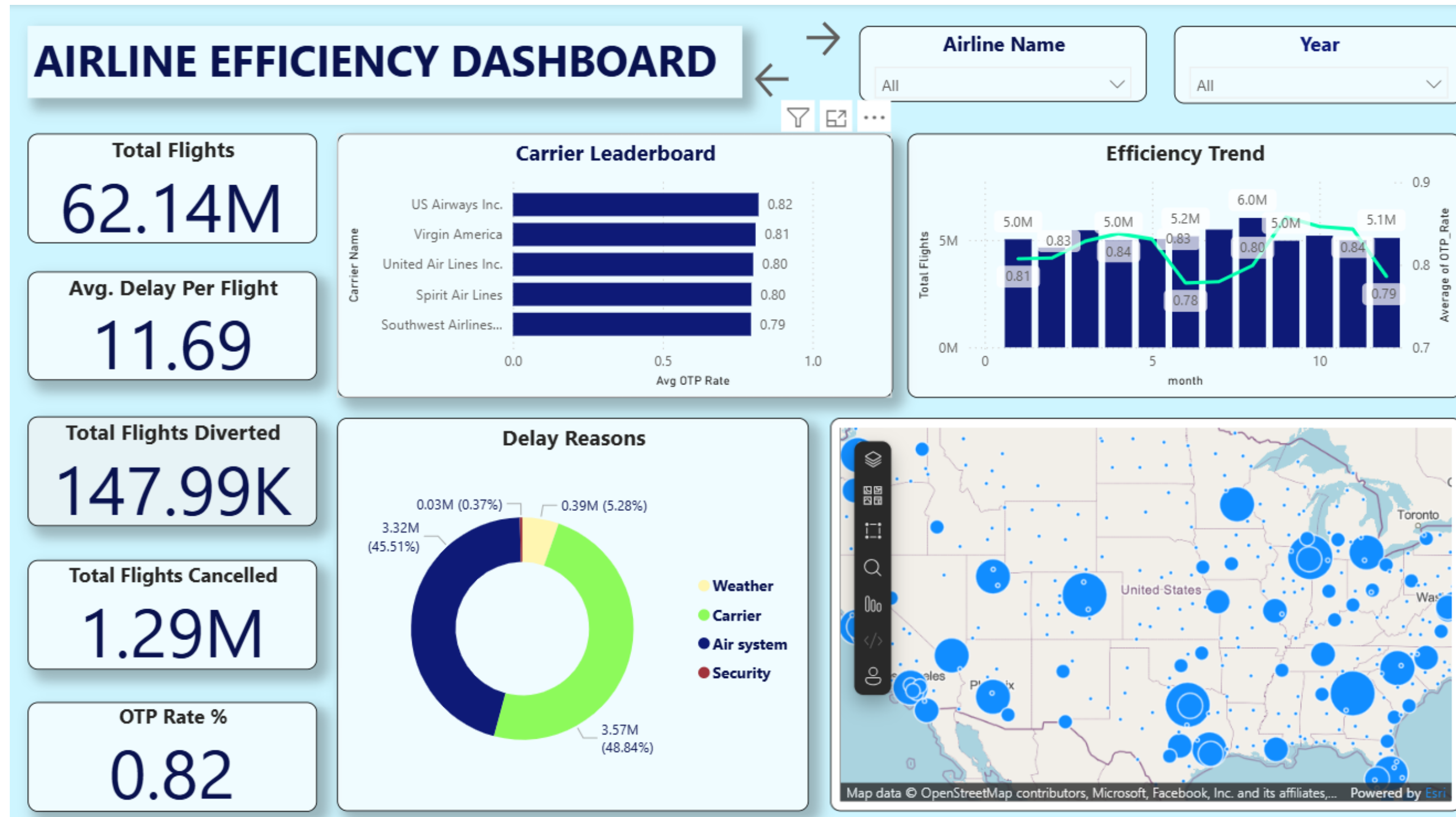
- Late Aircraft Delay: Confirmed as the #1 predictor, proving the "Ripple Effect" is the greatest threat to system stability.
- NAS Delay: Identified as the second most critical factor, highlighting the impact of Air Traffic Control congestion.
- Data-Driven Decisions: This shifts strategy away from uncontrollable factors (Weather) toward controllable operational bottlenecks (Turnaround Time).

DEPLOYMENT & ETL

Operationalizing Intelligence in the Data Warehouse

- The Predictive Fact Table: Created a specialized SQL table `fact_Predictive_Analytics` to store model outputs.
- Automated Ingestion: Used the SQLAlchemy engine to bridge Python intelligence with MySQL, replacing the table dynamically during each ETL run.
- Relational Mapping: Cleanly mapped Python predictions to `carrier_id`, `airport_id`, and `time_id` to maintain Star Schema integrity.
- BI Ready: This final step ensures the Power BI dashboard can slice "Predicted Risk" by Carrier or Date instantly.

POWER BI DASHBOARD



CONCLUSION

- End-to-End Pipeline: Successfully engineered a robust BI ecosystem processing 171,223 records from raw CSV to a structured MySQL Data Warehouse.
- Star Schema Implementation: Optimized data retrieval through a normalized architecture consisting of a central Fact_Delays table and supporting Dimension tables (Carrier, Airport, Time).
- Advanced Analytics Integration: Transitioned from descriptive reporting to predictive intelligence by deploying Random Forest and Z-Score models to identify system anomalies and operational risks.
- Dashboard Creation: Developed an Interactive Power BI Dashboard that serves as the visual interface for the DWH, translating complex ML outputs into intuitive executive scorecards and drill-down reports.



REFERENCE

1. Data Source (Primary)

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- Kimball, R., & Ross, M. (1996). The Data Warehouse Toolkit: The Complete Guide to Dimensional Modeling. John Wiley & Sons.

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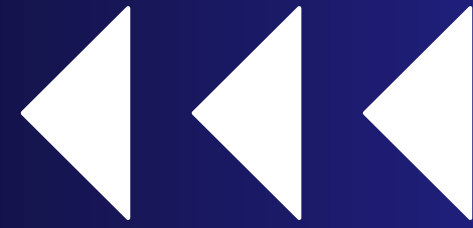
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- Few, S. (2006). Information Dashboard Design: The Effective Visual Communication of Data. O'Reilly Media.



THANK YOU